STOCHASTIC METHODS IN WATER RESOURCES

Unit 5: Geostatistics
Lecture 16: Generalities, descriptive spatial statistis

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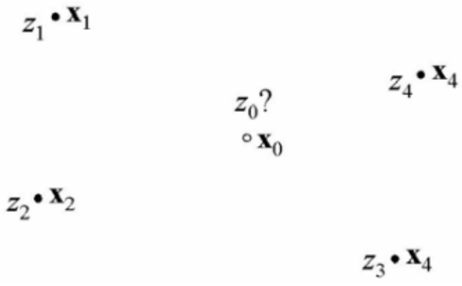
Generalities

- **Geostatistics** is the part of statistics that is concerned with geo-referenced data, i.e. data that are linked to spatial coordinates.
- To describe the spatial variation of the property observed at data locations, the property is modelled with a spatial random function (or random field) $Z(\mathbf{x})$, $\mathbf{x}^T = (x, y)$ or $\mathbf{x}^T = (x, y, z)$.
- The focus of geostatistics can be further explained by the figure below.



Suppose that the values of some property (for instance hydraulic conductivity) have been observed at the four locations x_1, x_2 , x_3 and x_4 and that, respectively, the values z_1 , z_2 , z_3 and z_4 have been found at these locations.

Generalities

- Geostatistics is concerned with the unknown value z_0 at the non-observed location x_0. In particular, geostatistics deals with:
  1. spatial interpolation and mapping: predicting the value of Z_0 at x_0 as accurately as possible, using the values found at the surrounding locations (note that Z_0 is written here in capitals to denote that it is considered to be a random variable);
  2. local uncertainty assessment: estimating the probability distribution of Z_0 at x_0 given the values found at the surrounding locations, i.e. estimating the probability density function $f_z\left(z_0; \mathbf{x}_0 | z_1(\mathbf{x}_1), z_2(\mathbf{x}_2), z_3(\mathbf{x}_3)\right)$. This probability distribution expresses the uncertainty about the actual but unknown value z_0 at $\mathbf{x}_0$;
  3. simulation: generating realisations of the conditional random function $Z(\mathbf{x}|z(\mathbf{x}_i), i = 1, \cdots, 4)$ at many non-observed locations $\mathbf{x}_i$ simultaneously (usually on a lattice or grid) given the values found at the observed locations; e.g. hydraulic conductivity is observed at a limited number of locations but must be input to a groundwater model on a grid.
- Geostatistics was first used as a practical solution to estimating ore grades of mining blocks (concentration or percentage of valuable minerals or metals within a block) using observations of ore grades that were sampled preferentially. i.e. along outcrops.Later it was extended to a comprehensive statistical theory for geo-referenced data.

Generalities

- Presently, geostatistics is applied in a great number of fields such as petroleum engineering, hydrology, soil science, environmental pollution and fisheries.
- Some important hydrological problems that have been tackled using geostatistics are among others:
  - spatial interpolation and mapping of rainfall depths and hydraulic heads;
  - estimation and simulation of representative conductivities of model blocks used in groundwater models;
  - simulation of subsoil properties such as rock types, texture classes and geological facies;
  - uncertainty analysis of groundwater flow and -transport through heterogeneous formations (if hydraulic conductivity, dispersivity or chemical properties are spatially varying and largely unknown)

Descriptive spatial statistics

Declustering

- Looking at figure above it can be seen that not all observation locations are evenly spread in space. Certain location appear to be clustered. This can for instance be the case because it is convenient to take a number of samples close together.
- Another reason could be that certain data clusters are taken purposively, e.g. to estimate the short distance variance.
- If the histogram or cumulative frequency distribution of the data is calculated with the purpose of estimating the true but unknown spatial frequency distribution of an area, it would not be fair to give the same weight to clustered observations as to observations that are far from the others. The latter represent a much larger area and thus deserve to be given more weight.
- To correct for the clustering effect declustering methods can be used. Here, one particular declustering method called **polygon declustering** is illustrated in the figure.
- The figure shows schematically a spatial array of measurement locations. The objective is to estimate the spatial statistics (mean, variance, histogram) of the property (e.g. hydraulic conductivity) of the field.

$$w_1 = \frac{A_1}{A_1 + A_2 + A_3 + A_4 + A_5}$$

$$w_2 = \frac{A_2}{A_1 + A_2 + A_3 + A_4 + A_5}$$

etc.

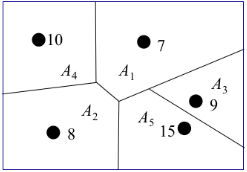
$A_1 = 0.30$

$A_2 = 0.25$

$A_3 = 0.10$

$A_4 = 0.20$

$A_5 = 0.15$



$$m_z = 0.3 \cdot 7 + 0.25 \cdot 8 + 0.10 \cdot 9 + 0.20 \cdot 10 + 0.15 \cdot 15 = 9.25$$

$$s_z^2 = 0.3 \cdot (-2.25)^2 + 0.25 \cdot (-1.25)^2 + 0.10 \cdot (-0.25)^2 + 0.20 \cdot (0.75)^2 + 0.15 \cdot (5.75)^2 = 6.99$$

Descriptive spatial statistics

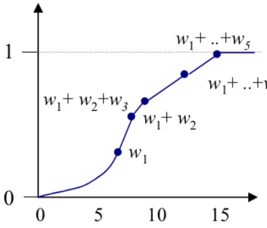
Declustering

- The idea is to draw **Thiessen polygons** around the observation locations first: by this procedure each location of the field is assigned to the closest observation.
- The relative sizes of the Thiessen polygons are used as declustering weights: $w_i = \frac{A_i}{\sum_j A_j}$.
- Using these weights the declustered histogram and cumulative frequency distribution can be calculated as shown in the figure below, as well as the declustered moments such as the mean and variance:

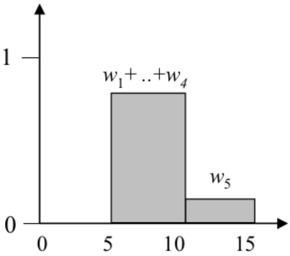
$$m_z = \sum_{i=1}^{n} w_i z_i \qquad (1)$$

$$s_z^2 = \sum_{i=1}^{n} w_i (z_i - m_z)^2 \qquad (2)$$

Declustered cum. Freq. Distr.

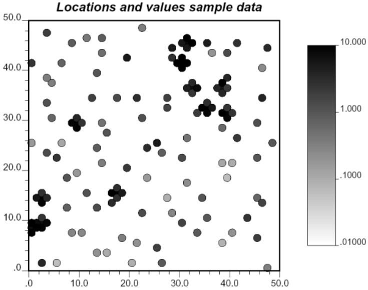


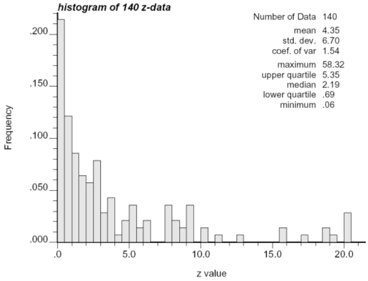
Declusterd histogram



Declustering

- The effect of declustering can be demonstrated using the synthetithe data-set shown in the figure below (all of the larger numerical examples shown in here are based on this data set).

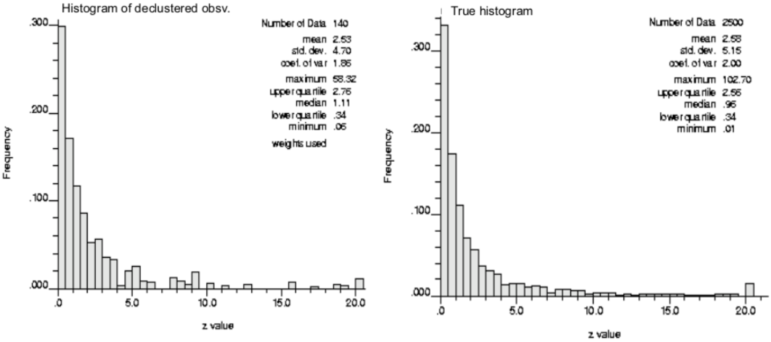




Descriptive spatial statistics

Declustering

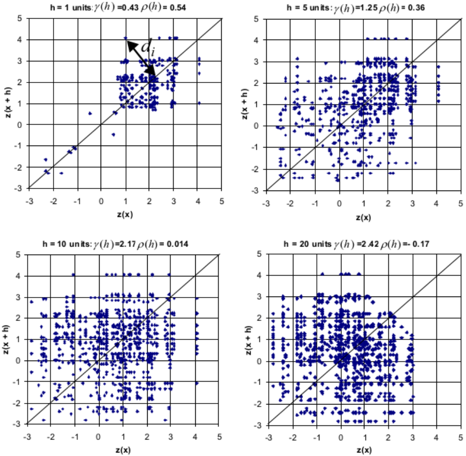
- The figure below shows the declusterd histogram based on the 140 data and the "true" histogram based on 2500 values.Clearly, the are very much alike, while the histogram based on the non-weighted data (see figure) is much different.



- The estimated mean without weighting equals 4.35 which is much too large. The reason is the existance of clusters with very high data values present in the observations (see figure above).
- Declustering can correct for this as can be seen from the declustered mean in the figure above which is 2.53 and very close to te true mean of 2.58.

Semivariance and correlation

- Using the synthetic data set of 140 observations we will further illustate the concept of the semivariogram and the correlation function.
- Figure below shows scatter plots of the $z(\mathbf{x})$ and $z(\mathbf{x}+\mathbf{h})$ for $|\mathbf{h}| = 1, 5, 10, 20$ units (pixels) apart. For each pair of points the distance d_i to the one-to-one line it can be calculated.



Semivariance and correlation

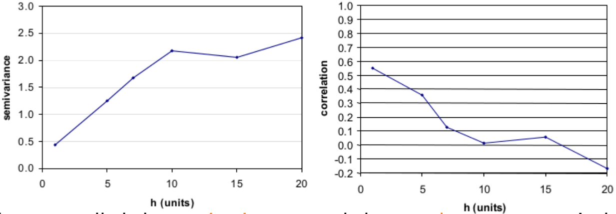
- The semivariance of a given distance is given by (with $n(h)$ the number of pairs of points that are a distance $h = |\mathbf{h}|$ apart)

$$\hat{\gamma}(h) = \frac{1}{2n(h)} \sum_{i=1}^{n(h)} d_i^2 = \frac{1}{2n(h)} \sum_{i=1}^{n(h)} [z(\mathbf{x}+\mathbf{h}) - z(\mathbf{x})]^2 \qquad (3)$$

  and the correlation coefficient

$$\hat{\rho}(h) = \frac{1}{n(h)} \sum_{i=1}^{n(h)} \frac{z(\mathbf{x}+\mathbf{h})z(\mathbf{x}) - m_{z(\mathbf{x}+\mathbf{h})} m_{z(\mathbf{x})}}{S_{z(\mathbf{x}+\mathbf{h})} S_{z(\mathbf{x})}} \qquad (4)$$

  $m_{z(\mathbf{x}+\mathbf{h})}$, $m_{z(\mathbf{x})}$ and $S_{z(\mathbf{x}+\mathbf{h})}$, $S_{z(\mathbf{x})}$ are the mean and variance of the $z(\mathbf{x}+\mathbf{h})$ and $z(\mathbf{x})$ data values respectively. The figure below shows plots of the semivariance and the correlation as a function of distance.



- These plots are called the semivariogram and the correlogram respectively. If we imagine the data z to be observations from a realisation of a random function $Z(\mathbf{x})$ and this random function is assumed to be intrinsic or wide sense stationary then equations 3 and 4 are estimators for the semivariance function and the correlation function.

Spatial interpolation by kriging

Generalities

- **Kriging** is a collection of methods that can be used for spatial interpolation.
- **Kriging** provides optimal linear predictions at non-observed locations by assuming that the unknown spatial variation of the property is a realisation of a random function that has been observed at the data points only.
- Apart from the prediction, kriging also provides the variance of the prediction
- Two kriging variants are discussed: **simple kriging** and **ordinary kriging** which are based on slightly different random function models.

Spatial interpolation by kriging

Simple kriging

The most elementary of kriging methods is called **simple kriging** and is based on a random function that is wide sense stationary, i.e. with the following properties

$$
\begin{aligned}
E[Z(\mathbf{x})] &= \mu_Z = \text{ constant} \\
var[Z(\mathbf{x})] &= E[Z(\mathbf{x}) - \mu_Z]^2 = \sigma_Z^2 = \text{ constant (and finite)} \\
COV\left[Z(\mathbf{x}_1), Z(\mathbf{x}_2)\right] &= E\left[(Z(\mathbf{x}_1) - \mu_Z)(Z(\mathbf{x}_2) - \mu_Z)\right] = C_Z(\mathbf{x}_2 - \mathbf{x}_1) = C(\mathbf{h})
\end{aligned}
$$

Simple kriging is the appropriate kriging method if the random forest is second order stationary and the mean of the random forest $E[Z(\mathbf{x})] = \mu$ is known without error. With simple kriging a predictor $\hat{Z}(\mathbf{x}_0)$ is sought that

1. is a linear function of the surrounding data,
2. is unbiased: $E[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = 0$,
3. and has the smallest possible error, i.e. $\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)$ is minimal.

A linear and unbiased predictor is obtained when considering the following weighted average of deviations from the mean:

$$
\hat{Z}(\mathbf{x}_0) = \mu_Z + \sum_{i=1}^{n} \lambda_i [Z(\mathbf{x}_i) - \mu_Z] \tag{5}
$$

with $Z(\mathbf{x}_i)$ the values of $Z(\mathbf{x})$ at the surrounding observation locations.

Spatial interpolation by kriging

Simple kriging

Usually, not all observed locations are included in the predictor, but only a limited number of locations within a given search neighbourhood. Equation 5 is unbiased by definition:

$$E[\hat{Z}(\mathbf{x}_0)-Z(\mathbf{x}_0)] = \mu_Z+\sum_{i=1}^{n}\lambda_i[Z(\mathbf{x}_i)-\mu_Z]-E[Z(\mathbf{x}_0)] = \mu_Z+\sum_{i=1}^{n}\lambda_i[\mu_Z-\mu_Z]-\mu_Z = 0 \tag{6}$$

The weights λ_i should be chosen such that the prediction error is minimal. However, as the real value $z(\mathbf{x}_0)$ is unknown, we cannot calculate the prediction error. Therefore, instead of minimizing the prediction error we must be satisfied with minimizing the variance of the prediction error $var[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)]$. Because the predictor is unbiased, the variance of the prediction error can be written as

$$\begin{aligned}
&var[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = E[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)]^2 = \\
&E\left[\sum_{i=1}^{n}\lambda_i E[Z(\mathbf{x}_i) - \mu_Z] - [Z(\mathbf{x}_0) - \mu_Z]\right]^2 = \\
&\sum_{i=1}^{n}\sum_{j=1}^{n}\lambda_i\lambda_j E\left[[Z(\mathbf{x}_i) - \mu_Z][Z(\mathbf{x}_j) - \mu_Z]\right] - \\
&2\sum_{i=1}^{n}\lambda_i E\left[[Z(\mathbf{x}_i) - \mu_Z][Z(\mathbf{x}_0) - \mu_Z]\right] + E[Z(\mathbf{x}_0) - \mu_z]^2
\end{aligned} \tag{7}$$

Spatial interpolation by kriging

Simple kriging

Using the definition of the covariance of a second order stationary stochastic field $E\left[[Z(\mathbf{x}_i) - \mu_Z][Z(\mathbf{x}_j) - \mu_Z]\right] = C_Z(\mathbf{x}_i - \mathbf{x}_j)$ and $C_Z(0) = \sigma_Z^2$, we obtain for the variance of the prediction error

$$var[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = \sum_{i=1}^{n}\sum_{j=1}^{n}\lambda_i\lambda_j C_Z(\mathbf{x}_i - \mathbf{x}_j) - 2\sum_{i=1}^{n}\lambda_i C_Z(\mathbf{x}_i - \mathbf{x}_0) + \sigma_Z^2 \qquad (8)$$

To obtain the minimum value of Equation 8 we have to equate all its partial derivatives with respect to the λ_i to zero:

$$\frac{\partial}{\partial\lambda_i} var[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = 2\sum_{j=1}^{n}\lambda_j C_Z(\mathbf{x}_i - \mathbf{x}_j) - 2C_Z(\mathbf{x}_i - \mathbf{x}_0) = 0 \qquad i = 1, 2, \cdots, n \quad (9)$$

This results in the following system of n equations referred to as the **simple kriging system**

$$\sum_{j=1}^{n}\lambda_j C_Z(\mathbf{x}_i - \mathbf{x}_j) = C_Z(\mathbf{x}_i - \mathbf{x}_0) \qquad i = 1, 2, \cdots, n \qquad (10)$$

The n unknown values λ_i can be uniquely solved from these n equations if all the $\mathbf{x}_i$ are different. The predictor in equation 5 with the λ_i found from solving equations 10 is the one with the minimum prediction error variance. This variance can be calculated using equation 8. However, it can be shown that the variance of the prediction error can be written in a simpler form as

$$var[\hat{Z}(\mathbf{x}_0) - Z(\mathbf{x}_0)] = \sigma_Z^2 - \sum_{i=1}^{n}\lambda_i C_Z(\mathbf{x}_i - \mathbf{x}_0) \qquad (11)$$

Spatial interpolation by kriging

Simple kriging

The error variance very nicely shows how kriging takes advantage of the spatial dependence of $Z(\mathbf{x}_i)$. If only the marginal probability distribution had been estimated from the data and the spatial coordinates had not been taken into account, the best prediction for every non-observed location would have been the mean μ_Z. Consequently, the variance of the prediction error would have been equal to σ_Z^2. As the larger kriging weights are positive, it can be seen from equation 13 that the prediction error variance of the kriging predictor is always smaller than the variance of the random function.

To obtain a positive error variance using equation 13 the function $C(\mathbf{h})$ must be positive definite. This means that for all possible $\mathbf{x}_1, \cdots, \mathbf{x}_n \in \Re^N$ ($N = 1, 2$ *or* 3) and for all $\lambda_1, \lambda_2, \cdots, \lambda_n \in \Re$ the following inequality must hold

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \lambda_i \lambda_j C_Z(\mathbf{x}_i - \mathbf{x}_j) > 0 \tag{12}$$

- It is difficult to ensure that this is the case for any covariance function. Therefore, we cannot just estimate a covariance function directly from the data for a limited number of separation distances and then obtain a continuous function by linear interpolation between the experimental points (such as in figure above).
- If such a covariance function were used in equation 10, equation 13 would not necessarily lead to a positive estimate of the prediction error variance.

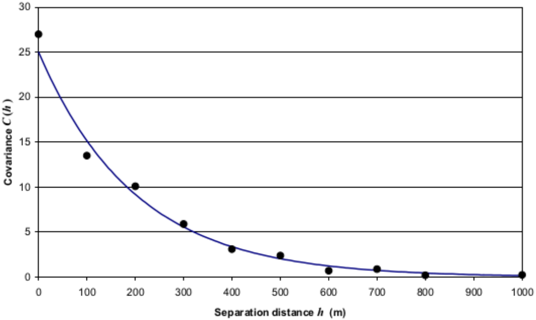
Simple kriging

- In fact, there are only a limited number of functions for which it is proven that inequality 12 will always hold. So the practical solution used in kriging is to take one of these 'permissible' functions and fit it through the points of the experimental covariance function.
- Next, the values of the fitted function are used to build the kriging system (see equation 10 and to estimate the kriging variance using equation 13.
- The table below gives a number of covariance functions that can be used for simple kriging (i.e. using a wide sense stationary random functions).Of course, in case of second order stationarity the parameter c should be equal to the variance of the random function: $c = \sigma_Z^2$.

Exponential covariance	$C_Z(h) = ce^{-h/a}$ for $h \geq 0$, $a > 0$
Gaussian covariance	$C_Z(h) = ce^{-h^2/a^2}$ for $h \geq 0$, $a > 0$
Spherical covariance	$C_Z(h) = \begin{cases} c\left[1 - \frac{3}{2}\left(\frac{h}{a}\right) + \frac{1}{2}\left(\frac{h}{a}\right)^3\right], \text{ if } h < a \\ 0 = , \text{ if } h \geq a, h \geq 0, a > 0 \end{cases}$
Nugget model	$\rho_Z(h) = \begin{cases} c, \text{ if } h = 0 \\ 0 = , \text{ if } h > 0 \end{cases}$

Simple kriging

- The figure below shows an example of an exponential model that is fitted to the estimated covariances.



- Any linear combination of a permissible covariance model is a permissible covariance model itself. Often a combination of a nugget model and another model is observed, e.g:

$$C(h) = \begin{cases} c_0 + c_1, \text{ if } h = 0 \\ c_1 e^{-h/a}, \text{ if } h > 0 \end{cases} \tag{13}$$

where $c_0 + c_1 = \sigma_Z^2$. In this case c_0 is often used to model the part of the variance that is attributable to observation errors or spatial variation that occurs at distances smaller than the minimal distance between observations.